

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks:20

Annexure A

Comparative Analysis

Code of Conduct:

I Moshitha Kolanjiyappan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Moshitha k

Annexure A

Short Answer Test

1, Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.

KNN and Case-Based Learning (CBL) are instance-based learning methods. For medical diagnosis, KNN stores patient records and classifies a new patient based on the majority class among the K most similar patients. CBL retrieves similar previous cases and reuses their diagnosis, possibly adapting it to the new patient.

Aspect	KNN	Case-Based Learning
Memory Usage	Stores training instances	Stores previous cases with solutions
Similarity	Distance measures such as Euclidean distance	Domain-specific similarity measures
Prediction	Majority voting or weighted voting	Retrieve, reuse, revise, and retain
Medical Example	Finds similar patients and predicts disease	Retrieves similar diagnosed patients and adapts diagnosis

Conclusion: KNN is simple and effective for classification, while CBL is more flexible when previous cases contain useful problem-solving knowledge.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

Consider a dataset containing temperature and machine-output values.

Aspect	Locally Weighted Regression	RBF Network
Locality	Builds a local model for each query	Uses radial basis functions around selected centers
Training	Very little/no explicit training	Requires training of centers and weights
Prediction Cost	High because weights are calculated for each query	Faster after training
Smoothness	Depends on bandwidth	Usually produces smooth predictions

Aspect	Locally Weighted Regression	RBF Network
Adaptability	Very high to local changes	Moderate after model training

Conclusion: LWR is useful when local patterns change significantly, while RBF networks are better when faster repeated predictions are required.

3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.

Suppose patients are classified as **Low Risk** or **High Risk** using age and blood pressure.

Aspect	KNN	RBF
Decision Boundary	Depends on nearby training points	Learned through radial basis functions
Training Time	Very low	Higher
Prediction Time	Higher for large datasets	Usually faster
Generalization	Depends strongly on K and data distribution	Generally smoother
Non-linearity	Naturally handles non-linear boundaries	Effectively models complex non-linear boundaries

Conclusion: KNN is easy to implement, while RBF networks are preferable when a smooth non-linear decision boundary and faster prediction are required.

4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

Consider a recommendation system that recommends products based on customer history.

Aspect	CBL	LWR
Data Handling	Stores previous cases	Uses nearby numerical observations
New Data	Retrieves and adapts similar cases	Gives higher weight to nearby data
Adaptability	Very high	High
Computation	Searching cases can be expensive	Recalculates local model
Output	Previous solution adapted to new case	Numerical prediction

Conclusion: CBL is suitable for recommendation systems involving reusable past experiences, while LWR is more suitable for predicting continuous values.

5. Compare Decision Tree Learning and K-Nearest Neighbour (KNN) using a classification problem such as customer churn prediction. Analyze differences in model interpretability, training time, handling of noisy data, and decision-making process. Discuss how each method performs with large datasets and complex feature spaces.

For customer churn prediction, features may include age, monthly bill, contract type, and usage.

Aspect	Decision Tree	KNN
Interpretability	Very high	Low
Training Time	Usually low	Very low
Prediction	Uses learned decision rules	Uses nearest customers
Noise Handling	Can overfit noise	Sensitive to noisy neighbors
Large Dataset	Efficient prediction	Computationally expensive
Feature Space	Handles mixed features well	Requires scaling

A Decision Tree provides rules such as **“If contract duration < 6 months and bill > ₹X, predict churn.”** KNN instead compares the customer with similar historical customers.

Conclusion: Decision Trees are generally preferable for large churn datasets because they are interpretable and faster during prediction.

6. Compare Naïve Bayes and Logistic Regression for a text classification task such as spam detection. Analyze assumptions about feature independence, probability estimation, computational efficiency, and performance on high-dimensional data. Discuss scenarios where one method outperforms the other.

Both algorithms can classify emails as spam or legitimate.

Aspect	Naïve Bayes	Logistic Regression
Assumption	Features are conditionally independent	No independence assumption
Output	Class probabilities	Class probabilities
Training	Very fast	Fast
High-Dimensional Data	Excellent	Excellent
Decision Boundary	Probabilistic	Linear
Text Classification	Very effective	Often highly effective

Naïve Bayes works particularly well with word-frequency features and small datasets. Logistic Regression can perform better when relationships between features are important.

Conclusion: Naïve Bayes is preferred for fast text classification, while Logistic Regression is useful when a discriminative decision boundary is desired.

7. Analyze the differences between Support Vector Machines (SVM) and KNN in classification tasks. Use an example dataset to compare decision boundaries, scalability, sensitivity to noise, and computational complexity. Discuss how kernel functions in SVM improve performance for non-linear problems.

Consider a dataset containing two classes of customer transactions.

Aspect	SVM	KNN
Decision Boundary	Learned during training	Determined during prediction
Scalability	Better for prediction	Expensive for large datasets
Noise Sensitivity	Depends on kernel and parameters	Sensitive to noisy neighbors
Training	Can be expensive	Minimal
Prediction	Generally fast	Can be expensive
Non-linear Data	Uses kernel functions	Naturally supports non-linear boundaries

SVM kernels such as RBF transform data into a space where a suitable separating boundary can be found.

Conclusion: SVM is generally better for large, high-dimensional classification problems, while KNN is simpler and useful for smaller datasets.

8. Compare K-Means Clustering and Hierarchical Clustering using a dataset with unknown structure. Analyze differences in cluster formation, computational cost, scalability, and interpretability. Discuss how each method handles varying cluster shapes and the impact of choosing the number of clusters.

Aspect	K-Means	Hierarchical Clustering
Cluster Formation	Centroid-based	Builds a hierarchy
Number of Clusters	Usually specified beforehand	Can be selected from dendrogram
Computational Cost	Relatively low	Higher
Scalability	Good for large datasets	Less suitable for very large datasets
Interpretability	Moderate	High using dendrogram
Cluster Shape	Best for roughly spherical clusters	Can reveal more complex structures

K-Means repeatedly assigns points to the nearest centroid. Hierarchical clustering progressively merges or divides clusters.

Conclusion: K-Means is efficient for large datasets, whereas hierarchical clustering is useful when the underlying cluster structure is unknown.

9. Compare Genetic Algorithms and Gradient Descent Optimization for solving optimization problems. Analyze differences in convergence behavior, ability to escape local minima, computational cost, and suitability for different types of objective functions. Provide examples where each method is more effective.

Aspect	Genetic Algorithm	Gradient Descent
Search	Population-based	Gradient-based
Local Minima	Better ability to escape	Can get trapped
Convergence	Usually slower	Usually faster
Computational Cost	High	Lower
Objective Function	Can work without derivatives	Requires differentiable objective
Best Use	Complex/discontinuous problems	Neural networks and regression

For example, Genetic Algorithms can optimize a complex scheduling problem, while Gradient Descent efficiently optimizes neural-network weights.

Conclusion: Genetic Algorithms are suitable for complex search spaces, whereas Gradient Descent is preferred for differentiable optimization problems.

10. Compare Principal Component Analysis (PCA) and Feature Selection techniques for dimensionality reduction. Analyze differences in information preservation, computational complexity, interpretability, and impact on model performance. Discuss how each approach affects downstream machine learning tasks.

Aspect	PCA	Feature Selection
Method	Creates new components	Selects original features
Information	Preserves maximum variance	May remove less useful information
Interpretability	Lower	High
Computation	Can be higher	Usually simpler
Features	Transformed	Original
Model	Can reduce dimensionality	Depends on selected features
Performance	effectively	

PCA combines correlated variables into principal components. Feature selection chooses the most relevant original variables.

Conclusion: PCA is useful when dimensionality is very high, while feature selection is preferred when interpretability of original features is important.

11. Compare Random Forest and Decision Tree algorithms using a classification problem such as credit risk assessment. Analyze differences in overfitting control, accuracy, interpretability, and robustness to noisy data. Discuss how ensemble learning in Random Forest improves generalization compared to a single decision tree.

For credit-risk assessment, both algorithms can classify customers as low-risk or high-risk.

Aspect	Decision Tree	Random Forest
Overfitting	More prone	Better controlled
Accuracy	Moderate to high	Usually higher
Interpretability	Very high	Lower
Noise	More sensitive	More robust
Training	Faster	More computationally expensive
Generalization	Lower	Better

Random Forest combines many decision trees using **ensemble learning**. Each tree works on different samples/features, and their predictions are combined.

Conclusion: Random Forest generally provides better accuracy and generalization, while a single Decision Tree is easier to interpret.

12. Compare Linear Regression and Polynomial Regression for predicting housing prices. Analyze differences in model complexity, ability to capture non-linear relationships, risk of overfitting, and interpretability. Discuss scenarios where each model performs better.

For housing-price prediction, house size, location, number of rooms, and age can be used as features.

Aspect	Linear Regression	Polynomial Regression
Model Complexity	Low	Higher
Relationship	Linear	Non-linear
Overfitting	Lower	Higher for large degrees
Interpretability	High	Lower
Computation	Low	Higher
Suitable For	Approximately linear trends	Curved relationships

Linear Regression assumes a straight-line relationship. Polynomial Regression adds powers of features to model curves.

Conclusion: Linear Regression is better for simple relationships, while Polynomial Regression is useful when housing prices show clear non-linear patterns.

13. Compare K-Means Clustering and DBSCAN using a dataset with irregular cluster shapes. Analyze differences in cluster detection, handling of noise and outliers, requirement of parameters, and scalability. Discuss how each algorithm performs in real-world clustering tasks.

Consider geographical data containing irregularly shaped groups of locations.

Aspect	K-Means	DBSCAN
Cluster Shape	Mainly spherical	Arbitrary shapes
Noise Handling	Poor	Excellent
Parameters	Number of clusters K	Epsilon and minimum points
Outliers	Assigned to clusters	Can be marked as noise
Scalability	Generally good	Good with suitable implementation
Number of Clusters	Must be specified	Automatically identifies clusters

DBSCAN groups points based on density and can identify outliers.

Conclusion: DBSCAN is more suitable for irregular cluster shapes and noisy real-world datasets, while K-Means is simpler and faster for compact clusters.

14. Compare Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for time-series prediction tasks such as stock price forecasting. Analyze differences in handling long-term dependencies, vanishing gradient problem, computational complexity, and prediction accuracy.

For stock-price or time-series prediction:

Aspect	RNN	LSTM
Short-Term Dependencies	Good	Good
Long-Term Dependencies	Poorer	Better
Vanishing Gradient	More affected	Reduced using gates
Complexity	Lower	Higher
Training	Faster	Slower
Accuracy	Good for simple sequences	Better for long dependencies

LSTM uses **input, forget, and output gates** with a memory cell to preserve important information over long sequences.

Conclusion: RNNs are suitable for simpler sequential patterns, while LSTMs are generally better when long-term dependencies are important.

15. Compare Naïve Bayes and Support Vector Machines (SVM) for classification problems such as sentiment analysis. Analyze differences in assumptions, decision boundaries, computational efficiency, and performance on high-dimensional data. Discuss scenarios where each method is preferable.

For sentiment analysis, text can be represented using word frequencies or TF-IDF features.

Aspect	Naïve Bayes	SVM
Assumption	Conditional feature independence	No independence assumption
Decision Boundary	Probabilistic	Maximum-margin boundary
Training	Very fast	Usually more computationally expensive
High-Dimensional Data	Excellent	Excellent
Text Classification	Very effective	Often highly accurate
Probability	Naturally provides probabilities	Requires calibration for probabilities

Naïve Bayes is useful when fast training and simple probabilistic classification are required. SVM is preferable when classification accuracy and a strong separating boundary are more important.

Conclusion: Naïve Bayes is suitable for fast, large-scale text classification, while SVM is often preferred when maximum classification performance is required.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand